BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Quiz - 1 | Semester: Spring 2026
Duration: 25 minutes | Full Marks: 20

CSE 340: Computer Architecture

Name: Solution	ID:	Section:

1.

Instruction Set:

Add ....
Sub ....
Ld ....
Add ....
Sd ....
Sub ....
Sll ....
Add ....
Sub ....
Sub ....
Ld ....
Sd ....
Sub ....
Sll ....

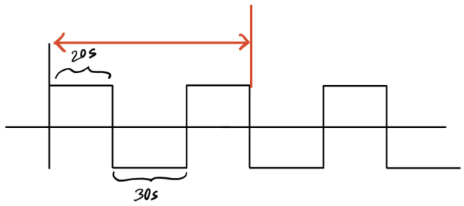

Figure : Sll Instruction

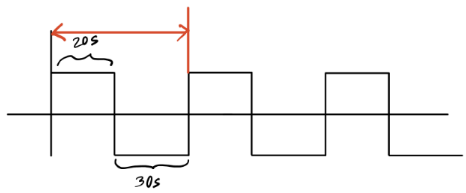

Figure : Ld Instruction

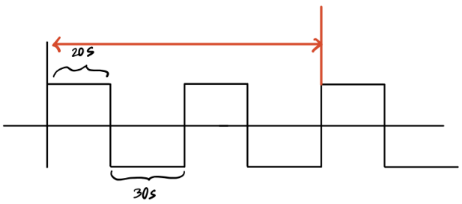

Figure : Sd Instruction

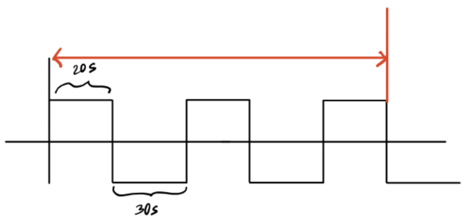

Figure : Sub Instruction

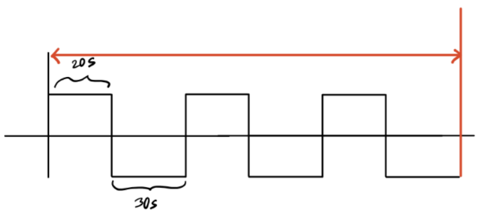

Figure : Add Instruction

a. Find CPIs; You must answer in the following table. [1 x 5]

	Add	Sub	Ld	Sd	Sll
CPI	3	2.5	1	2	1.5
IC	3	5	2	2	2

b. Find the Frequency in **GHz**. Answer: __________ [1]

c. Find the time to run the instruction set. [4]

c)

$$\text{Avg. CPI} = \frac{(3\times 3) + (2.5\times 5) + (1\times 2) + (2\times 2) + (1.5\times 2)}{14} = \frac{30.5}{14} = 2.17$$

CPU Time = $14 \times 2.17 \times 50 = 1525$s

b)

Clock Cycle Time $= (30+20)$s
$= 50$s

$$\text{Freq.} = \frac{1}{\text{Clock Cycle Time}}$$

$$= \frac{1}{50}\ \text{Hz}$$

$$= 0.02 \times 10^{-9}\ \text{GHz}$$

2. A hospital server that manages patient records is equipped with dual power supplies. If one power supply fails due to a surge, the other continues to provide power, preventing any downtime or loss of service.

Write the name of the relevant idea from the 7 Great Idea(s) that is applied in the given scenario. [2]

Answer: Dependability via Redundancy.

3. A program runs in 40 seconds on a reference computer. The SPEC ratio of Computer A for this program is 0.80. Detailed profiling of Computer A shows that 65% of the total execution time of the program is spent executing ADD instructions. You plan to improve the performance of Computer A by enhancing only the ADD instruction , while the execution time of all other instructions remains unchanged.

a) Find the execution time of the program on Computer A. [2]

b) Determine how much the ADD instruction must be sped up so that Computer A matches the execution time of the reference computer. [6] → improvement factor!

Ref. time

Soln

a) $\text{Spec Ratio} = \frac{\text{Ref. time}}{\text{CPU time}}$

$\Rightarrow \text{CPU time} = \frac{\text{Ref. time}}{\text{Spec Ratio}}$

$= \frac{40}{0.80}$

$= 50s$

Ref. time $= 40s$

Spec Ratio $= 0.80$

New time = Ref. time $= 40s$

Old time $= 50s$

$T_{aff.} = (50 \times 0.65) = 32.5s$

$T_{unaff.} = (50 - 32.5) = 17.5s$

b) $T_{new} = \frac{T_{aff}}{n} + T_{unaff}$

$\Rightarrow 40 = \frac{32.5}{n} + 17.5$

$\Rightarrow \frac{32.5}{n} = 22.5$

$\Rightarrow n = 1.44$